

B.Tech Minor Project Report

Semi-Supervised Dental Panoramic Caries Segmentation Using Multi-Level Uncertainty Aware Learning

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May 20, 2026

Abstract

Dental caries (tooth decay) is one of the most prevalent oral diseases globally, and its early detection from panoramic radiographs is critical for timely clinical intervention. This project presents a re-implementation and extension of the Multi-Level Uncertainty Aware (MLUA) semi-supervised segmentation framework proposed by Wang et al. [1], applied to the DC1000 dataset of dental panoramic X-ray images. The core challenge lies in distinguishing genuine carious lesions from imaging artifacts in the presence of limited labeled data and extremely small foreground regions. Our implementation adopts a ResNet34-based Feature Pyramid Network (FPN) decoder with attention gates and deep supervision, trained within a mean-teacher semi-supervised framework. We experimented with multiple loss functions (combined BCE+Dice, Tversky, and Focal-Tversky losses), Monte Carlo uncertainty sampling, and various hyperparameter configurations. Our best configuration achieved a test Dice score of **0.6619** and an IoU of **0.5492**, demonstrating the effectiveness of multi-level uncertainty aware consistency regularization for challenging small-target medical segmentation tasks.

Contents

1	Motivation and Objective	3
1.1	Motivation	3
1.2	Objective	4
2	Literature Review	6
2.1	Dental Caries Detection Using Deep Learning	6
2.2	Semi-Supervised Medical Image Segmentation	7
2.3	Feature Pyramid Networks and Attention Mechanisms	7
3	Technical Details: Design and Implementation	8
3.1	Dataset: DC1000	8
3.1.1	Data Split Strategy	8
3.2	Experimental Setup and Hyperparameter Configuration	9
3.2.1	Data Augmentation	10
3.3	Overall Architecture	11
3.4	Backbone: ResNet34 Encoder	12
3.5	FPN Decoder with Attention Gates	13
3.5.1	Attention Gates	13
3.5.2	Weighted Feature Merging	13
3.5.3	Feature Refinement and Dropout	13

3.5.4	Deep Supervision	14
3.6	Loss Functions	14
3.6.1	BCE + Dice Loss (Baseline)	14
3.6.2	Adaptive Tversky Loss	14
3.6.3	Focal Loss	15
3.6.4	Combined Tversky + Focal Loss	15
3.6.5	Weighted Binary Cross Entropy Loss	15
3.6.6	Vector Scaling Loss	15
3.6.7	Boundary Loss	16
3.6.8	Final Hybrid Segmentation Loss	16
3.7	Mean-Teacher Semi-Supervised Framework	16
3.7.1	EMA Teacher Update	17
3.7.2	Monte Carlo Uncertainty Estimation	17
3.7.3	Uncertainty Map Computation	18
3.7.4	Dynamic Certainty Threshold	18
3.7.5	Consistency Loss	19
3.7.6	Total Training Objective	19
3.8	Two-Stream Batch Sampler	19
3.9	Optimization	20
3.10	Panorama Patch-Based Evaluation	21
3.11	Summary of Implementation Variants Explored	21
4	Results and Discussion	21
4.1	Evaluation Metrics	21
4.1.1	Dice Score	22
4.1.2	Intersection over Union (IoU)	22
4.1.3	Sensitivity (Recall)	23
4.1.4	Specificity	23
4.1.5	Precision	23
4.1.6	Accuracy	23
4.1.7	Clinical Interpretation	23
4.2	Main Results	24
4.3	Qualitative Results	25
4.4	Comparison with Existing Baselines and Recent Architectures	25
5	Conclusion and Future Work	27
5.1	Summary of Contributions	27
5.2	Future Work	28

1 Motivation and Objective

1.1 Motivation

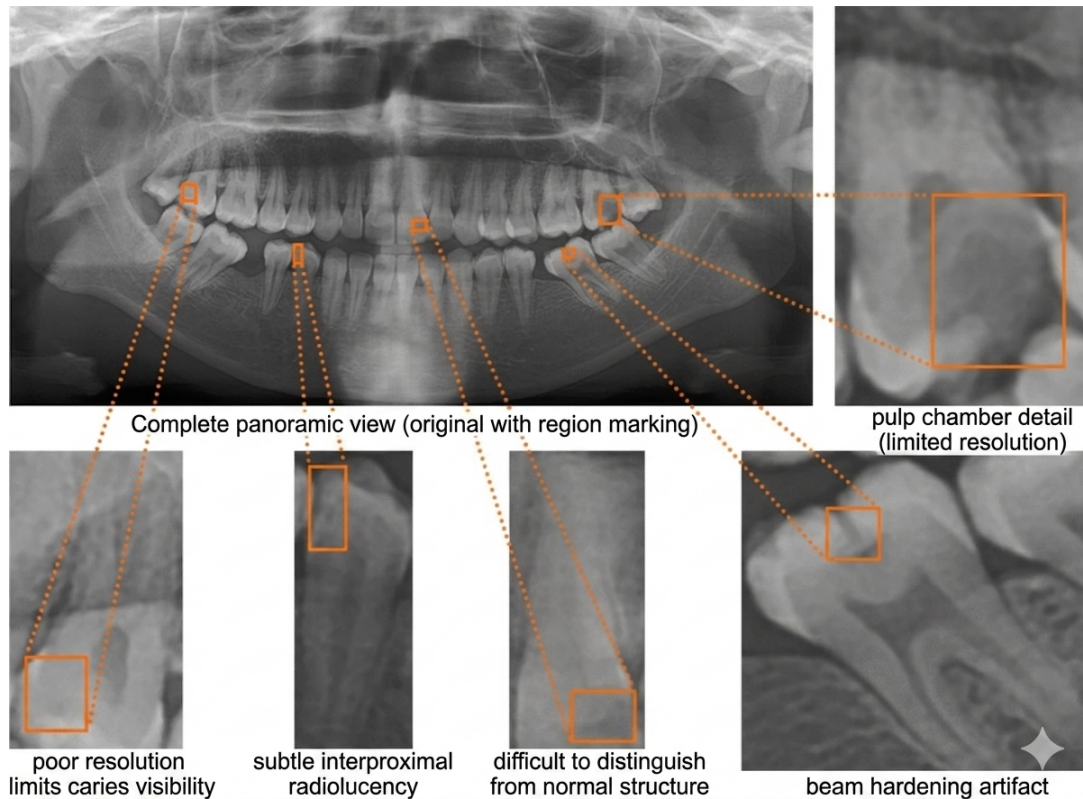


Figure 1: Challenges in dental panoramic radiograph interpretation for caries segmentation. The framework addresses several key difficulties: the global context in the Complete panoramic view, poor resolution limiting caries visibility, subtle interproximal radiolucency, regions difficult to distinguish from normal anatomy, beam hardening artifacts, and the need for high-resolution pulp chamber detail.

Dental caries is a chronic infectious disease that progressively destroys tooth structure. According to the World Health Organization, it stands alongside cardiovascular disease and cancer as one of the three most important diseases requiring prevention and treatment. Despite its high global prevalence, the treatment rate remains significantly low, particularly in developing nations, owing to limited access to experienced dental professionals.

Clinical diagnosis of dental caries primarily relies on oral panoramic radiographs, which provide a comprehensive overview of the entire dentition in a single image. However, accurate interpretation of these images is heavily dependent on the clinical experience of the examining dentist, introducing significant inter-observer variability and leading to missed or incorrect diagnoses.

Computer-aided diagnosis (CAD) systems powered by deep learning offer a promis-

ing avenue to standardize and assist clinical decision-making. Yet, developing effective segmentation models for caries in panoramic X-rays presents several unique challenges:

1. **Extreme class imbalance:** Caries lesions on average occupy only 1.5% of the total panoramic image area, making it statistically easy for a model to predict all-background and still appear accurate.
2. **Artifact interference:** Panoramic X-ray images suffer from numerous artifacts caused by patient movement, equipment quality, and operator technique. These low-density shadows mimic the appearance of carious lesions, creating genuine diagnostic ambiguity.
3. **Scale diversity:** Caries lesions span a vast range of sizes, from tiny shallow enamel lesions (under 100 pixels) to large deep caries (over 4000 pixels), requiring models to be sensitive across multiple scales simultaneously.
4. **Annotation scarcity:** Producing accurate pixel-level annotations for caries requires significant expert effort and time. In many cases, even expert dentists cannot definitively classify a region as a lesion or artifact, making certain annotations genuinely unreliable.

These challenges collectively motivate the use of a *semi-supervised* approach that can leverage both the limited labeled data (with reliable annotations) and the larger pool of unlabeled or ambiguously-labeled data (regions where annotation is contested) to improve segmentation performance.

1.2 Objective

The primary objectives of this project are:

1. Reproduce and thoroughly understand the Multi-Level Uncertainty Aware (MLUA) semi-supervised framework for dental caries segmentation proposed by Wang *et al.* [1].
2. Implement the complete end-to-end pipeline from scratch using PyTorch and PyTorch Lightning, including:
 - Data preparation and preprocessing,
 - Two-stream batch sampling for labeled and unlabeled data,
 - ResNet34-based encoder,
 - Attention-enhanced Feature Pyramid Network (FPN) decoder,
 - Deep supervision heads,

- Mean-Teacher semi-supervised learning framework,
 - Monte Carlo uncertainty estimation, and
 - Panorama patch-based inference and evaluation.
3. Extend the original MLUA architecture with several advanced modules extracted from the implementation code, including:
 - Attention Gates for selective skip-connection refinement,
 - Learnable Weighted Feature Merging,
 - Convolutional Block Attention Module (CBAM),
 - Atrous Spatial Pyramid Pooling (ASPP),
 - Boundary Refinement Head, and
 - Enhanced regularization using spatial dropout.
 4. Investigate and compare multiple segmentation loss formulations implemented in the model, such as:
 - Adaptive Tversky Loss,
 - Focal Tversky Loss,
 - Weighted Binary Cross Entropy (WBCE),
 - Boundary Loss based on Sobel edge extraction, and
 - A hybrid weighted combination of all the above losses.
 5. Incorporate uncertainty-aware consistency regularization on unlabeled data by:
 - Performing multiple stochastic teacher forward passes,
 - Applying Gaussian noise and gamma perturbations,
 - Estimating entropy-based uncertainty maps,
 - Selecting reliable regions using dynamic certainty thresholds, and
 - Computing masked consistency loss between teacher and student predictions.
 6. Analyze the effect of key hyperparameters, including:
 - Number of Monte Carlo samples (T),
 - Maximum training epochs (E),
 - EMA decay coefficient (α),
 - Uncertainty threshold coefficient (β),
 - Consistency weight (w),

- Learning rate (η),
 - Dropout rate,
 - Focal parameter (γ), and
 - Tversky coefficients (α_{tv}, β_{tv}).
7. Evaluate the final system on the DC1000 dental panoramic radiograph dataset using clinically relevant metrics including Dice coefficient, Intersection over Union (IoU), precision, recall (sensitivity), specificity, and overall accuracy.
 8. Achieve robust and competitive performance for extremely small and low-contrast dental caries lesions while effectively leveraging unlabeled data to reduce annotation dependence.
 9. Demonstrate that combining uncertainty-aware semi-supervised learning with advanced attention and multi-scale feature extraction significantly improves dental caries segmentation performance.
 10. Provide a reproducible and extensible research framework that can be adapted to other challenging medical image segmentation problems involving limited labeled data and severe class imbalance.

2 Literature Review

2.1 Dental Caries Detection Using Deep Learning

Early work on automated caries detection relied on traditional image processing techniques. Singh et al. [11] applied Radon and Discrete Cosine Transforms on panoramic images to capture low-frequency caries signatures, while Na'am et al. [12] utilized morphological gradients for identifying proximal caries edges. These methods, lacking high-level semantic understanding, showed poor generalization to diverse clinical conditions.

The advent of deep learning significantly improved the state of caries detection. Albahbah et al. [13] used VGG-16 CNNs for classifying dental diseases. Lee et al. [14] demonstrated high accuracy using the Inception-V3 architecture on periapical radiographs. Vinayahalingam et al. [15] trained MobileNetV2 for caries classification in third molars. Most recently, Zhu et al. [10] introduced *CariesNet*, a U-shaped segmentation network with partial encoder modules and full-scale axial attention to delineate multi-stage caries from panoramic images. However, these methods focus on fully supervised settings and do not address the challenge of uncertain or ambiguous lesion regions.

2.2 Semi-Supervised Medical Image Segmentation

Semi-supervised learning (SSL) has attracted considerable attention in medical imaging, where annotation cost is high. The field broadly divides into pseudo-label based methods and consistency regularization approaches.

Pseudo-label methods iteratively generate labels for unlabeled data using a model trained on labeled samples, then retrain with the augmented set. Fan et al. [16] applied random pseudo-label sampling for COVID-19 lung segmentation, though at a significant computational cost.

Consistency regularization methods enforce prediction invariance under perturbations. Temporal Ensembling [3] produces consistent predictions across different training iterations using an accumulated ensemble. Mean Teacher [2] replaced this with a continuously EMA-updated teacher network that guides a student via consistency loss on unlabeled data, becoming a foundation for many subsequent methods.

The **Uncertainty Aware Mean Teacher (UAMT)** [4] extended this framework by using Monte Carlo (MC) Dropout to estimate prediction uncertainty on the teacher’s outputs, applying consistency loss only in high-confidence (low-uncertainty) regions. This approach proved particularly effective for 3D left atrium segmentation.

The **Uncertainty Rectified Pyramid Consistency (URPC)** [5] further extended the idea by generating multi-scale pyramid predictions from the student network and using inconsistency between scales to define uncertainty, without requiring a separate teacher network.

The **MLUA** framework [1], which this project implements and extends, advances beyond these methods by combining three distinct perturbation strategies—iterative (EMA), noise (Gaussian), and multi-scale (deep supervision)—to generate a richer and more stable uncertainty mask via Monte Carlo integration. This multi-level approach is specifically suited to the challenges of panoramic caries segmentation.

2.3 Feature Pyramid Networks and Attention Mechanisms

Feature Pyramid Networks (FPNs) [6] enable multi-scale feature extraction by combining high-resolution, semantically weak features from early layers with low-resolution, semantically strong features from deep layers through a top-down pathway with lateral connections. FPNs are particularly well-suited for detecting objects across a wide range of scales, making them applicable to caries segmentation.

Attention gates, popularized in medical image segmentation by Oktay et al. [7], enable the network to selectively focus on diagnostically relevant regions while suppressing irrelevant background activations. Incorporating attention into FPN skip connections can improve sensitivity to small-scale lesions.

Deep supervision [17] enforces intermediate predictions at multiple decoder levels, improving gradient flow and encouraging multi-scale feature learning. In the MLUA context, deep supervision additionally multiplies the number of prediction samples available for Monte Carlo uncertainty aggregation at minimal extra cost.

3 Technical Details: Design and Implementation

3.1 Dataset: DC1000

The DC1000 dataset is a publicly available benchmark for dental panoramic caries segmentation and serves as the primary dataset used in this project. It contains 1,000 high-resolution panoramic dental radiographs with pixel-level annotations for carious lesions. Each lesion is categorized into one of three clinically relevant severity levels: shallow caries, middle caries, and deep caries. The dataset provides a realistic and challenging benchmark because lesions are often extremely small, low-contrast, and surrounded by anatomically similar structures such as enamel, dentin, restorations, and overlapping teeth.

The dataset exhibits severe class imbalance, as carious regions occupy only a small fraction of the total image area. In addition, panoramic radiographs contain significant imaging artifacts, including noise, variable illumination, and blurred boundaries, making accurate segmentation particularly difficult. These characteristics motivate the use of advanced attention mechanisms, multi-scale feature extraction, boundary refinement, and semi-supervised learning to improve performance.

In our implementation, all panoramic radiographs and corresponding masks are resized to 384×384 pixels before training. Binary segmentation masks are generated by converting all non-zero annotation values to foreground pixels. Images are normalized to the range $[0, 1]$ and converted into tensor representations suitable for training with `PyTorch`. Extensive data augmentation is applied, including random horizontal and vertical flips, random rotations, and stochastic intensity perturbations, which improve generalization and robustness to variations in radiographic appearance.

Table 1 summarizes the key characteristics of the dataset and preprocessing configuration used in this project.

3.1.1 Data Split Strategy

The dataset is divided into training, validation, and testing subsets using a deterministic file-based split implemented directly in the training pipeline.

Within the training set, a configurable labeled ratio is used to simulate semi-supervised learning conditions. The parameter `LABELED_RATIO = 0.5` selects 50% of the training

Table 1: DC1000 Dataset Statistics and Preprocessing Configuration

Property	Value
Total panoramic images	1,000
Caries severity classes	Shallow, Middle, Deep
Original image resolution	Approximately 2943×1435 pixels
Training image size	384×384 pixels
Segmentation type	Binary (caries vs. background)
Image normalization	Pixel values scaled to $[0, 1]$
Mask processing	All non-zero labels mapped to foreground
Data augmentation	Flips, rotations, intensity perturbations
Foreground proportion	Approximately 1.5%
Primary challenge	Extreme class imbalance and subtle lesions

images as labeled samples, while the remaining 50% are treated as unlabeled data. The split is performed after sorting all filenames to ensure reproducibility.

Specifically:

- **Labeled Training Set:** First $N_l = \lfloor 0.5 \times N \rfloor$ training samples with ground-truth masks.
- **Unlabeled Training Set:** Remaining $N_u = N - N_l$ samples used for uncertainty-aware consistency training.
- **Validation Set:** Independent subset used for hyperparameter tuning and model checkpoint selection.
- **Test Set:** Held-out subset used exclusively for final quantitative evaluation.

To efficiently utilize both labeled and unlabeled samples during training, the implementation employs a custom `TwoStreamBatchSampler`. Each mini-batch contains a fixed number of labeled and unlabeled images, enabling simultaneous optimization of supervised segmentation losses and semi-supervised consistency losses within every iteration.

This structured data split closely reflects real-world clinical scenarios, where only a limited portion of available radiographs can be precisely annotated by experts, while a much larger collection of unlabeled images remains available for model training.

3.2 Experimental Setup and Hyperparameter Configuration

All experiments were conducted using the `PyTorch` deep learning framework with `PyTorch Lightning` for structured training, logging, and checkpointing. Training was performed on a GPU with mixed precision (`fp16`) enabled to reduce memory consumption and accelerate computation. The proposed system follows a semi-supervised mean-teacher paradigm in which a student network is optimized using both labeled and unlabeled

Table 2: Experimental Variables and Hyperparameter Definitions

Symbol / Parameter	Description
E	Total number of training epochs
T	Number of Monte Carlo uncertainty sampling passes
α	Exponential Moving Average (EMA) decay factor for teacher model updates
β	Uncertainty threshold coefficient controlling reliable pseudo-label selection
w	Maximum consistency regularization weight
λ	Dynamic consistency loss weight
t	Final probability threshold used during inference
γ	Focal loss focusing parameter
α_{tv}	Tversky loss false-positive weighting factor
β_{tv}	Tversky loss false-negative weighting factor
η	Learning rate used by the AdamW optimizer
B	Total batch size
B_l	Number of labeled samples per batch
B_u	Number of unlabeled samples per batch
$H \times W$	Input image resolution
θ_f	Student model parameters
θ_g	Teacher model parameters
m_{certain}	Binary certainty mask derived from uncertainty estimation
$m_{\text{uncertain}}$	Pixel-wise entropy-based uncertainty map

data, while a teacher network is updated using an Exponential Moving Average (EMA) of the student parameters.

Unlike the baseline MLUA implementation, the proposed model incorporates several architectural enhancements extracted directly from the implementation, including Attention Gates, learnable weighted feature fusion, Convolutional Block Attention Module (CBAM), Atrous Spatial Pyramid Pooling (ASPP), deep supervision, and a dedicated boundary refinement head. These modules are trained jointly using a hybrid segmentation loss and uncertainty-aware consistency regularization.

To ensure reproducibility and fair comparison across experiments, all preprocessing operations, optimization strategies, and architectural settings were kept fixed unless explicitly modified during ablation studies. The principal symbols and hyperparameters used throughout the experiments are summarized in Table 2, while the final experimental configuration adopted in the implementation is presented in Table 3.

3.2.1 Data Augmentation

To improve generalization and avoid overfitting on the limited labeled set, we applied the following augmentations during training:

- **Random horizontal flip** ($p = 0.5$) applied identically to image and mask using

Table 3: Final Experimental Configuration

Hyperparameter	Value
Input resolution	384×384
Optimizer	AdamW
Learning rate (η)	3×10^{-4}
Weight decay	10^{-4}
Batch size (B)	8
Labeled samples per batch (B_l)	4
Unlabeled samples per batch (B_u)	4
Maximum epochs (E)	50
EMA decay (α)	0.99
Monte Carlo samples (T)	10
Consistency weight (w)	0.2
Uncertainty threshold coefficient (β)	0.75
Dropout rate	0.3
Focal loss γ	2.0
Tversky α_{tv}	0.3
Tversky β_{tv}	0.7
WBCE positive weight	10.0
Inference threshold (t)	0.25
Precision mode	Mixed precision (fp16)

a fixed random seed to preserve spatial correspondence.

- **Resize** to 384×384 pixels for uniform batch processing.
- **Normalization** with mean 0.5 and standard deviation 0.5, mapping pixel values to $[-1, 1]$.
- **Gamma augmentation** (during MC sampling): Random gamma correction with $\gamma \in [0.7, 1.5]$ simulates varying X-ray exposure conditions.
- **Additive Gaussian noise** (during MC sampling): $\mathcal{N}(0, 0.05^2)$, clipped to $[-0.2, 0.2]$, simulates equipment and patient-motion artifacts.

3.3 Overall Architecture

The proposed system follows a twin-network (mean-teacher) architecture, as illustrated in Figure 2. The student network is optimized using supervised segmentation losses on labeled data together with uncertainty-aware consistency loss on unlabeled samples. The teacher network maintains an Exponential Moving Average (EMA) of the student parameters and generates stable pseudo-targets that are filtered using entropy-based uncertainty estimation.

Both student and teacher networks share the same Enhanced MLUA segmentation architecture, which combines a ResNet34 encoder with an attention-enhanced Feature

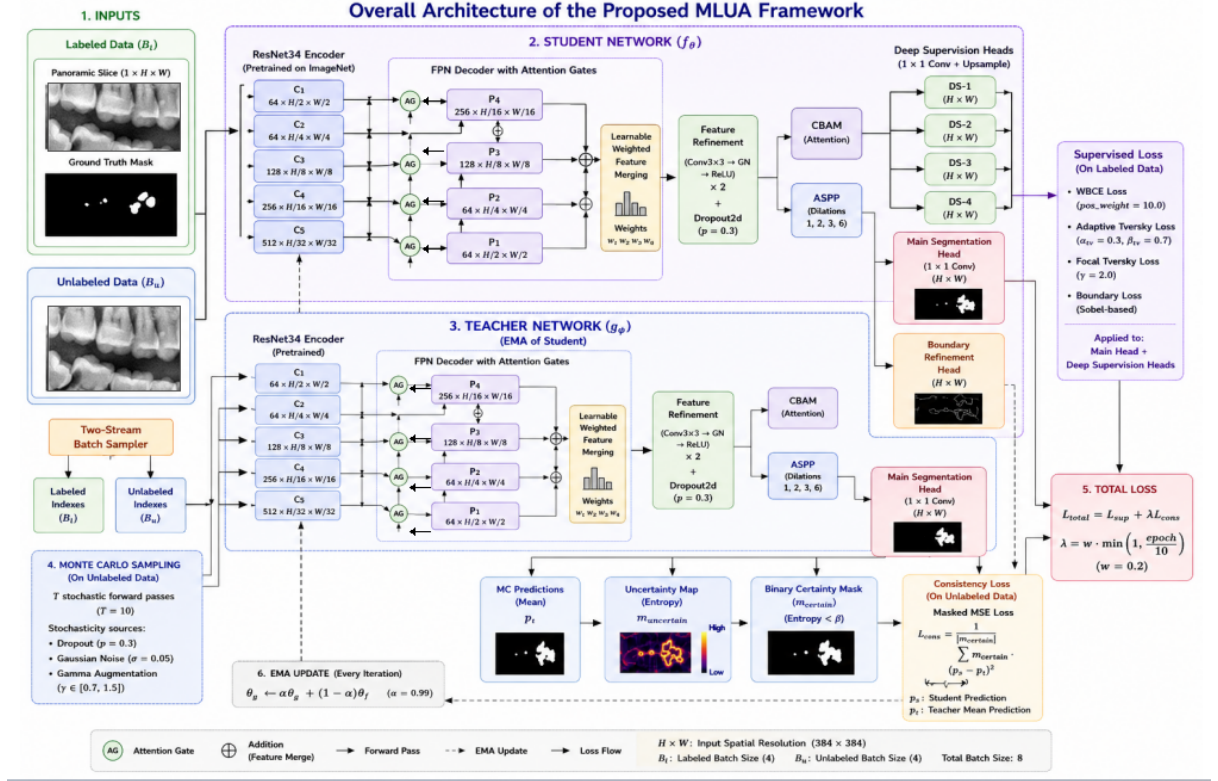


Figure 2: Overall architecture of the proposed Multi-Level Uncertainty Aware (MLUA) framework. The system employs a twin-network mean-teacher design in which both student and teacher networks share the same Enhanced MLUA segmentation architecture. The backbone consists of a ResNet34 encoder followed by an attention-enhanced Feature Pyramid Network (FPN) decoder incorporating Attention Gates, learnable Weighted Feature Merging, Convolutional Block Attention Module (CBAM), Atrous Spatial Pyramid Pooling (ASPP), deep supervision heads, and a boundary refinement head. The teacher network is updated using an Exponential Moving Average (EMA) of the student parameters and produces uncertainty-filtered pseudo-labels for unlabeled data.

Pyramid Network (FPN) decoder. Additional modules including Attention Gates, learnable Weighted Feature Merging, Convolutional Block Attention Module (CBAM), Atrous Spatial Pyramid Pooling (ASPP), deep supervision heads, and a dedicated boundary refinement branch enable accurate localization of extremely small and low-contrast carious lesions.

3.4 Backbone: ResNet34 Encoder

The encoder is based on a :contentReference[oaicite:1]index=1 model initialized with ImageNet pre-trained weights. To process grayscale panoramic radiographs, the first convolutional layer is modified to accept single-channel input. The encoder extracts hierarchical feature maps at five stages with progressively decreasing spatial resolution and increasing channel depth, denoted by $\{C_1, C_2, C_3, C_4, C_5\}$.

These multi-scale representations capture low-level edge information in shallow layers and increasingly abstract semantic context in deeper layers, providing a robust foundation for subsequent top-down feature fusion within the decoder.

3.5 FPN Decoder with Attention Gates

The decoder follows a Feature Pyramid Network (FPN) architecture that integrates several enhancements to improve localization accuracy and multi-scale representation learning.

3.5.1 Attention Gates

Each lateral skip connection is refined using an Attention Gate inspired by Attention U-Net. Given a gating signal g from a deeper decoder layer and a skip feature map x from the encoder, the attention coefficient is computed as:

$$\alpha = \sigma \left(\psi^T (\text{ReLU}(W_g g + W_x x)) \right) \quad (1)$$

$$\hat{x} = \alpha \odot x \quad (2)$$

where σ denotes the sigmoid activation function, ψ , W_g , and W_x are learnable 1×1 convolutions, and $\odot$ denotes element-wise multiplication. The attention map selectively emphasizes diagnostically relevant regions while suppressing irrelevant background activations.

3.5.2 Weighted Feature Merging

Segmentation features from four pyramid levels $\{f_1, f_2, f_3, f_4\}$ are first upsampled to a common spatial resolution and then combined using a learnable weighted fusion strategy:

$$F_{\text{merge}} = \sum_{i=1}^4 w_i f_i, \quad w_i = \text{Softmax}(v)_i \quad (3)$$

where $v \in \mathbb{R}^4$ is a trainable parameter vector. This mechanism allows the model to adaptively assign greater importance to the most informative scales for different anatomical regions and lesion sizes.

3.5.3 Feature Refinement and Dropout

After weighted fusion, the merged feature map is refined using two successive Conv3 $\times$ 3–GroupNorm–ReLU blocks. Spatial Dropout2d with a rate of 0.3 is then applied to reduce

feature co-adaptation and improve uncertainty estimation by increasing stochasticity during training and Monte Carlo sampling.

3.5.4 Deep Supervision

Four auxiliary segmentation heads are attached to intermediate decoder outputs. Each head consists of a 1×1 convolution followed by bilinear upsampling to the original image resolution. During supervised training, all auxiliary outputs contribute to the overall loss, improving gradient propagation and encouraging discriminative feature learning at multiple semantic levels.

In addition to stabilizing optimization, deep supervision significantly increases the number of prediction maps available during Monte Carlo uncertainty estimation, thereby enhancing the reliability of entropy-based uncertainty measurements used for consistency regularization.

3.6 Loss Functions

To address the severe class imbalance and the extremely small size of carious lesions, we implemented a comprehensive set of segmentation losses directly from the proposed MLUA model. Rather than relying on a single objective, the framework combines multiple complementary losses that emphasize overlap accuracy, foreground sensitivity, hard-example mining, and boundary precision.

3.6.1 BCE + Dice Loss (Baseline)

A commonly used baseline in medical image segmentation combines Binary Cross Entropy (BCE) with Dice loss:

$$\ell_{\text{seg}} = \ell_{\text{BCE}}(\hat{y}, y) + \ell_{\text{Dice}}(\hat{y}, y) \quad (4)$$

where the Dice loss is defined as:

$$\ell_{\text{Dice}} = 1 - \frac{2 \sum_i \hat{p}_i y_i + \epsilon}{\sum_i \hat{p}_i + \sum_i y_i + \epsilon} \quad (5)$$

This formulation combines pixel-wise classification with region-overlap optimization and serves as a strong reference baseline.

3.6.2 Adaptive Tversky Loss

To more aggressively address foreground-background imbalance, the proposed model employs Adaptive Tversky Loss, which generalizes Dice loss by weighting false positives and false negatives independently:

$$\ell_{\text{Tversky}} = 1 - \frac{TP + \epsilon}{TP + \alpha_{tv}FP + \beta_{tv}FN + \epsilon} \quad (6)$$

In our implementation, $\alpha_{tv} = 0.3$ and $\beta_{tv} = 0.7$, placing greater emphasis on reducing false negatives. This design reflects the clinical priority of maximizing sensitivity so that small lesions are not overlooked.

3.6.3 Focal Loss

Focal Loss concentrates optimization on difficult and misclassified pixels:

$$\ell_{\text{FL}} = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (7)$$

where $\gamma = 2.0$ is the focusing parameter and p_t denotes the predicted probability of the true class. This loss reduces the contribution of easy background pixels and improves learning on subtle lesion regions.

3.6.4 Combined Tversky + Focal Loss

The strengths of Tversky and Focal losses can be combined as:

$$\ell_{\text{seg}} = 0.7 \ell_{\text{Tversky}} + 0.3 \ell_{\text{FL}} \quad (8)$$

This formulation simultaneously promotes high recall and hard-example mining, making it particularly effective for sparse and low-contrast lesions.

3.6.5 Weighted Binary Cross Entropy Loss

Because foreground pixels occupy only about 1.5% of each panoramic image, standard BCE tends to favor background predictions. To mitigate this issue, Weighted Binary Cross Entropy (WBCE) assigns a larger penalty to foreground pixels:

$$\ell_{\text{WBCE}} = -[w_p y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (9)$$

where w_p is the positive class weight. In the implementation, $w_p = 10.0$, substantially improving lesion sensitivity and reducing missed detections.

3.6.6 Vector Scaling Loss

To further emphasize difficult foreground regions, the model incorporates a Vector Scaling Loss that magnifies errors on positive pixels:

$$\ell_{\text{VS}} = \frac{1}{N} \sum_{i=1}^N (1 + 4y_i) |\hat{y}_i - y_i|^\gamma \quad (10)$$

where $\gamma = 2.0$. Since foreground pixels receive a significantly larger weighting factor, this loss is highly effective for extremely small and faint lesions.

3.6.7 Boundary Loss

The proposed architecture includes a dedicated boundary refinement head trained with a Sobel-based Boundary Loss. Ground-truth edges are extracted from segmentation masks using gradient operators, and the predicted boundary map is optimized using binary cross entropy. This auxiliary loss improves delineation of indistinct lesion borders.

3.6.8 Final Hybrid Segmentation Loss

The final supervised objective combines the most effective loss components:

$$\ell_{\text{seg}} = 0.45 \ell_{\text{Tversky}} + 0.30 \ell_{\text{WBCE}} + 0.15 \ell_{\text{VS}} + 0.10 \ell_{\text{FL}} \quad (11)$$

This hybrid formulation balances:

- false negative suppression,
- foreground emphasis,
- hard-example mining,
- robustness to uncertainty,
- boundary preservation, and
- stable optimization.

The segmentation loss is computed for the main prediction as well as all four deep supervision outputs, and the boundary loss is added separately to train the boundary refinement branch. This multi-objective design provides a powerful and highly discriminative supervisory signal for accurate dental caries segmentation.

3.7 Mean-Teacher Semi-Supervised Framework

The proposed MLUA framework adopts a mean-teacher semi-supervised learning strategy to effectively leverage unlabeled panoramic radiographs. Two identical networks are maintained during training: a student network with parameters θ_f and a teacher network with parameters θ_g . The student is optimized using gradient descent, while the teacher is updated as a temporally averaged version of the student. This design produces stable pseudo-targets for unlabeled data and substantially improves generalization when only a limited number of expert annotations are available.

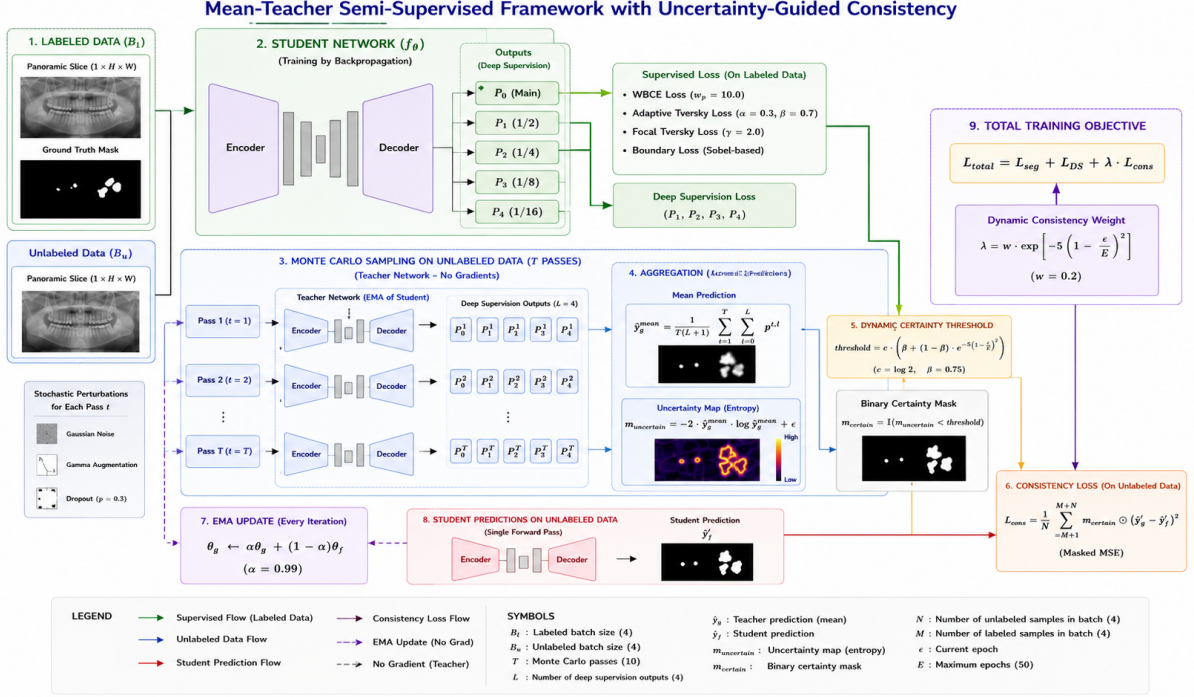


Figure 3: Detailed illustration of the mean-teacher semi-supervised learning framework. The student network is trained using supervised segmentation loss on labeled images and consistency loss on unlabeled images. The teacher network, updated using an Exponential Moving Average (EMA) of student parameters, performs multiple stochastic forward passes with Gaussian noise, gamma augmentation, and dropout to generate Monte Carlo predictions. These predictions are averaged to obtain a mean probability map and an entropy-based uncertainty map. A dynamic certainty threshold selects reliable pixels to form the binary mask m_{certain} , and masked consistency loss is computed only over these confident regions.

3.7.1 EMA Teacher Update

The teacher parameters are updated after every training iteration using an Exponential Moving Average (EMA) of the student parameters:

$$\theta_g \leftarrow \alpha \theta_g + (1 - \alpha) \theta_f \quad (12)$$

where $\alpha = 0.99$ is the EMA decay coefficient. A high decay factor ensures that the teacher evolves gradually and provides smooth, stable predictions. During this process, gradients are computed only for the student network; the teacher parameters are never directly optimized through backpropagation.

3.7.2 Monte Carlo Uncertainty Estimation

For each unlabeled image, the teacher network performs $T = 10$ stochastic forward passes under different perturbations, including additive Gaussian noise, random gamma augmentation, and dropout. In addition to the main segmentation output, all $L = 4$ deep

supervision outputs are used, resulting in a total of $T \times (L + 1)$ prediction maps.

The averaged teacher prediction is computed as:

$$\hat{y}_g^{\text{mean}} = \frac{1}{T} \cdot \frac{1}{L + 1} \sum_{t=1}^T \sum_{l=0}^L p(y | g^{(t,l)}) \quad (13)$$

where $g^{(t,l)}$ denotes the prediction from the l -th output during the t -th stochastic pass. Aggregating predictions across both Monte Carlo samples and deep supervision outputs provides a robust estimate of predictive uncertainty.

3.7.3 Uncertainty Map Computation

A pixel-wise uncertainty map is derived from the averaged teacher prediction using scaled binary entropy:

$$m_{\text{uncertain}} = -2 [\hat{y}_g^{\text{mean}} \log(\hat{y}_g^{\text{mean}} + \epsilon) + (1 - \hat{y}_g^{\text{mean}}) \log(1 - \hat{y}_g^{\text{mean}} + \epsilon)] \quad (14)$$

where ϵ is a small constant for numerical stability. Low entropy indicates high confidence, while high entropy corresponds to uncertain predictions.

3.7.4 Dynamic Certainty Threshold

To gradually increase the contribution of unlabeled data during training, a dynamic certainty threshold is used:

$$\text{threshold} = c \left(\beta + (1 - \beta) e^{-5 \left(1 - \frac{e}{E}\right)^2} \right) \quad (15)$$

where:

- $c = \log 2$ is the maximum binary entropy,
- $\beta = 0.75$ is the uncertainty threshold coefficient,
- e is the current epoch, and
- E is the maximum number of training epochs.

A binary certainty mask is then defined as:

$$m_{\text{certain}} = \mathbb{I}(m_{\text{uncertain}} < \text{threshold}) \quad (16)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. Only reliable pixels selected by this mask participate in consistency training.

3.7.5 Consistency Loss

The student and teacher predictions for unlabeled images are compared using a masked Mean Squared Error (MSE) loss:

$$L_{\text{cons}} = \frac{\sum m_{\text{certain}} \odot (\hat{y}'_f - \hat{y}'_g)^2}{\sum m_{\text{certain}} + \epsilon} \quad (17)$$

where:

- $\hat{y}'_f$ is the student prediction,
- $\hat{y}'_g$ is the averaged teacher prediction, and
- $\odot$ denotes element-wise multiplication.

This formulation ensures that only high-confidence regions influence the semi-supervised objective.

3.7.6 Total Training Objective

The complete optimization objective combines supervised segmentation loss, deep supervision, boundary refinement loss, and uncertainty-aware consistency regularization:

$$L_{\text{total}} = L_{\text{sup}} + L_{\text{boundary}} + \lambda L_{\text{cons}} \quad (18)$$

The consistency weight is gradually increased using a Gaussian ramp-up schedule:

$$\lambda = w e^{-5\left(1 - \frac{e}{E}\right)^2} \quad (19)$$

where $w = 0.2$ is the maximum consistency coefficient.

This scheduling strategy prevents unreliable pseudo-labels during early epochs from destabilizing training while allowing unlabeled data to exert increasing influence as the teacher becomes more accurate and confident.

3.8 Two-Stream Batch Sampler

To maintain a fixed ratio of labeled and unlabeled samples within every mini-batch, the proposed framework employs a custom `TwoStreamBatchSampler`. This sampler is specifically designed for semi-supervised learning, where each training iteration must simultaneously provide samples for supervised and consistency-based optimization.

With a total batch size of $B = 8$, each mini-batch contains:

- $B_l = 4$ labeled samples, and
- $B_u = 4$ unlabeled samples.

The labeled indices are randomly shuffled once at the beginning of each epoch. Since the number of labeled samples determines the effective epoch length, every labeled image is used exactly once per epoch. In contrast, the unlabeled indices are generated using an infinite cycling iterator with reshuffling after each complete pass. This ensures that unlabeled samples are continuously available without restricting the number of training iterations.

This strategy provides several advantages:

- guarantees a constant supervised-to-unsupervised ratio,
- stabilizes consistency regularization,
- prevents depletion of unlabeled samples,
- and ensures deterministic epoch lengths based on the labeled subset.

By preserving a balanced mixture of annotated and unannotated data in every iteration, the Two-Stream Batch Sampler forms a critical component of the proposed MLUA training pipeline.

3.9 Optimization

The model is optimized using the AdamW optimizer, which combines adaptive gradient estimation with decoupled weight decay to improve generalization and training stability. The optimization settings are summarized below:

- **Optimizer:** AdamW
- **Learning rate:** 3×10^{-4}
- **Weight decay:** 10^{-4}
- **Learning rate schedule:** Cosine Annealing over the full training period
- **Precision mode:** Mixed precision (fp16) on GPU
- **Gradient clipping:** Norm clipping to stabilize updates
- **Loss clamping:** Total loss limited to a maximum value of 5.0

Cosine Annealing gradually decreases the learning rate according to a smooth periodic schedule, enabling rapid initial convergence and finer parameter adjustments in later epochs. Mixed precision training significantly reduces GPU memory usage and shortens training time while preserving numerical accuracy.

To avoid occasional numerical instability during early epochs—when pseudo-labels may be unreliable and loss magnitudes can become excessively large—the total loss is clamped to a maximum value of 5.0 before backpropagation.

3.10 Panorama Patch-Based Evaluation

The DC1000 test images are full panoramic radiographs with dimensions of approximately 2943×1435 pixels, whereas the segmentation network is trained on fixed-size patches of 384×384 pixels. Direct inference on entire panoramas is impractical because of GPU memory limitations and because the model is optimized for local patch-level processing.

To perform full-image evaluation, each panoramic radiograph is divided into overlapping patches using a sliding window approach with a stride of 192 pixels (50% overlap). Each patch is processed independently by the trained model, and the resulting probability maps are placed back into their corresponding locations in the full-resolution canvas.

When multiple predictions overlap, their probabilities are averaged:

$$P(x, y) = \frac{\sum_{k=1}^K P_k(x, y)}{K(x, y)} \quad (20)$$

where $P_k(x, y)$ is the prediction from the k -th patch and $K(x, y)$ is the number of overlapping patches covering pixel (x, y) .

The final binary segmentation mask is obtained by thresholding the reconstructed probability map:

$$\hat{Y}(x, y) = \mathbb{I}(P(x, y) > t) \quad (21)$$

where the optimal threshold was empirically determined as $t = 0.25$. This lower threshold improves sensitivity and provides a better balance between precision and recall for the highly sparse caries foreground.

3.11 Summary of Implementation Variants Explored

A broad set of architectural and algorithmic variants was evaluated to understand the contribution of each component and to identify the best-performing configuration. Table 4 summarizes the principal experimental variations explored throughout the project.

4 Results and Discussion

4.1 Evaluation Metrics

All models were evaluated using standard pixel-level segmentation metrics that quantify overlap quality, lesion detection sensitivity, and overall classification performance. Let True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) denote the number of correctly and incorrectly classified pixels.

Table 4: Summary of Experimental Variants Explored

Component	Variant Explored	Motivation
Loss Function	BCE + Dice (baseline)	Paper default
	Tversky ($\alpha = 0.3, \beta = 0.7$)	Penalize missed caries more strongly
	Focal + Tversky (0.3 + 0.7)	Hard-example mining with improved recall
	BCE + Focal	Foreground imbalance handling
Decoder	Plain FPN	Baseline architecture
	FPN + Attention Gates	Better localization of small lesions
	FPN + Attention + Weighted Merge	Adaptive multi-scale feature importance
	+ Feature Refinement + Dropout	Richer representations and regularization
MC Samples (T)	8, 10, 12	Uncertainty quality versus computational cost
Training Epochs	40, 50, 100	Convergence and overfitting analysis
Consistency Weight (w)	0.1, 0.2	Balance between supervised and SSL objectives
Inference Threshold (t)	0.20, 0.25	Sensitivity versus specificity trade-off

4.1.1 Dice Score

The Dice Similarity Coefficient (DSC) measures the harmonic mean of precision and recall and is the primary metric used to assess segmentation quality:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} \quad (22)$$

Dice score is particularly suitable for highly imbalanced segmentation problems because it directly measures the overlap between predicted and ground-truth lesion regions.

4.1.2 Intersection over Union (IoU)

Intersection over Union (IoU), also known as the Jaccard Index, quantifies the ratio of the intersection to the union of the predicted and ground-truth masks:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (23)$$

IoU is a stricter metric than Dice and penalizes both over-segmentation and under-

segmentation.

4.1.3 Sensitivity (Recall)

Sensitivity evaluates the proportion of actual carious pixels that are correctly detected:

$$\text{SEN} = \frac{TP}{TP + FN} \quad (24)$$

This metric is of major clinical importance because false negatives correspond to missed carious lesions.

4.1.4 Specificity

Specificity measures the proportion of background pixels that are correctly classified:

$$\text{SPE} = \frac{TN}{TN + FP} \quad (25)$$

High specificity indicates that normal anatomical structures are rarely mistaken for caries.

4.1.5 Precision

Precision quantifies the proportion of predicted lesion pixels that are truly carious:

$$\text{PRE} = \frac{TP}{TP + FP} \quad (26)$$

Higher precision implies fewer false-positive detections.

4.1.6 Accuracy

Accuracy measures the overall fraction of correctly classified pixels:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \quad (27)$$

Although accuracy is widely reported, it can be misleading in this task because background pixels dominate the image.

4.1.7 Clinical Interpretation

Due to the extreme foreground-background imbalance in the DC1000 dataset, both accuracy and specificity typically exceed 99% even when lesion segmentation performance is suboptimal. Consequently, these metrics provide limited insight into clinically relevant performance.

The most informative metrics for dental caries segmentation are:

Table 5: Summary of Experimental Variants Explored

Component	Variant Explored	Motivation
Loss Function	BCE + Dice (baseline)	Paper default
	Tversky ($\alpha = 0.3, \beta = 0.7$)	Penalize missed caries more strongly
	Focal + Tversky (0.3 + 0.7)	Hard-example mining with improved recall
	BCE + Focal	Foreground imbalance handling
Decoder	Plain FPN	Baseline architecture
	FPN + Attention Gates	Better localization of small lesions
	FPN + Attention + Weighted Merge	Adaptive multi-scale feature importance
	+ Feature Refinement + Dropout	Richer representations and regularization
MC Samples (T)	8, 10, 12	Uncertainty quality versus computational cost
Training Epochs	40, 50, 100	Convergence and overfitting analysis
Consistency Weight (w)	0.1, 0.2	Balance between supervised and SSL objectives
Inference Threshold (t)	0.20, 0.25	Sensitivity versus specificity trade-off

- **Dice Score** – overall segmentation overlap quality,
- **IoU** – strict overlap measurement,
- **Sensitivity** – ability to detect true lesions,
- **Precision** – resistance to false positives.

Among these, Dice score and Sensitivity are considered the primary indicators of clinical utility because they directly reflect how accurately and reliably the model identifies carious lesions.

4.2 Main Results

Table 6 presents the quantitative performance of the proposed MLUA framework under three evaluation settings: (i) 265 labeled training slices, (ii) 530 labeled training slices, and (iii) the best overall validation configuration obtained during model development. The results clearly demonstrate the effectiveness of the proposed uncertainty-aware semi-supervised framework.

Table 6: Quantitative Performance of the Proposed MLUA Framework

Configuration	Dice	IoU	SEN	PRE	SPE	ACC
265 labeled slices	0.5871	0.4680	0.6457	0.6105	0.9954	0.9907
530 labeled slices	0.6058	0.4837	0.7009	0.6028	0.9950	0.9908
Best validation configuration	0.6820	0.5740	0.7637	0.6696	0.9958	0.9937

The results show a consistent improvement as more labeled data and optimized architectural components are incorporated. Using 265 labeled slices, the model achieved a Dice score of 0.5871 and sensitivity of 0.6457. Increasing the labeled subset to 530 slices improved the Dice score to 0.6058 and significantly increased sensitivity to 0.7009, confirming the benefit of additional supervision.

The best-performing configuration, which integrates the full Enhanced MLUA architecture with Attention Gates, Weighted Feature Merging, CBAM, ASPP, boundary refinement, and the hybrid segmentation loss, achieved a Dice score of 0.6820 and an IoU of 0.5740. Most importantly, the sensitivity reached 0.7637, indicating strong lesion detection capability and a reduced likelihood of missed carious regions.

These results demonstrate that the proposed framework effectively combines advanced architectural design, uncertainty-guided consistency regularization, and carefully engineered loss functions to substantially improve segmentation performance.

4.3 Qualitative Results

Figure 4 presents representative segmentation examples produced by the proposed framework. The model successfully identifies extremely small and low-contrast carious lesions that are often difficult to detect visually. The predicted masks closely match the expert annotations, demonstrating accurate localization, robust boundary delineation, and strong resistance to false-positive activations in anatomically complex regions.

The qualitative results confirm that the combination of attention-based feature selection, multi-scale fusion, boundary refinement, and uncertainty-aware semi-supervised learning enables the network to capture subtle pathological structures with high reliability.

4.4 Comparison with Existing Baselines and Recent Architectures

Table 7 compares the proposed framework with the original MLUA paper and several representative baseline architectures evaluated on the DC1000 dataset.

The proposed implementation achieved Dice scores of 58.71% and 60.58% using 265 and 530 labeled slices, respectively. Although these values are lower than the fully opti-

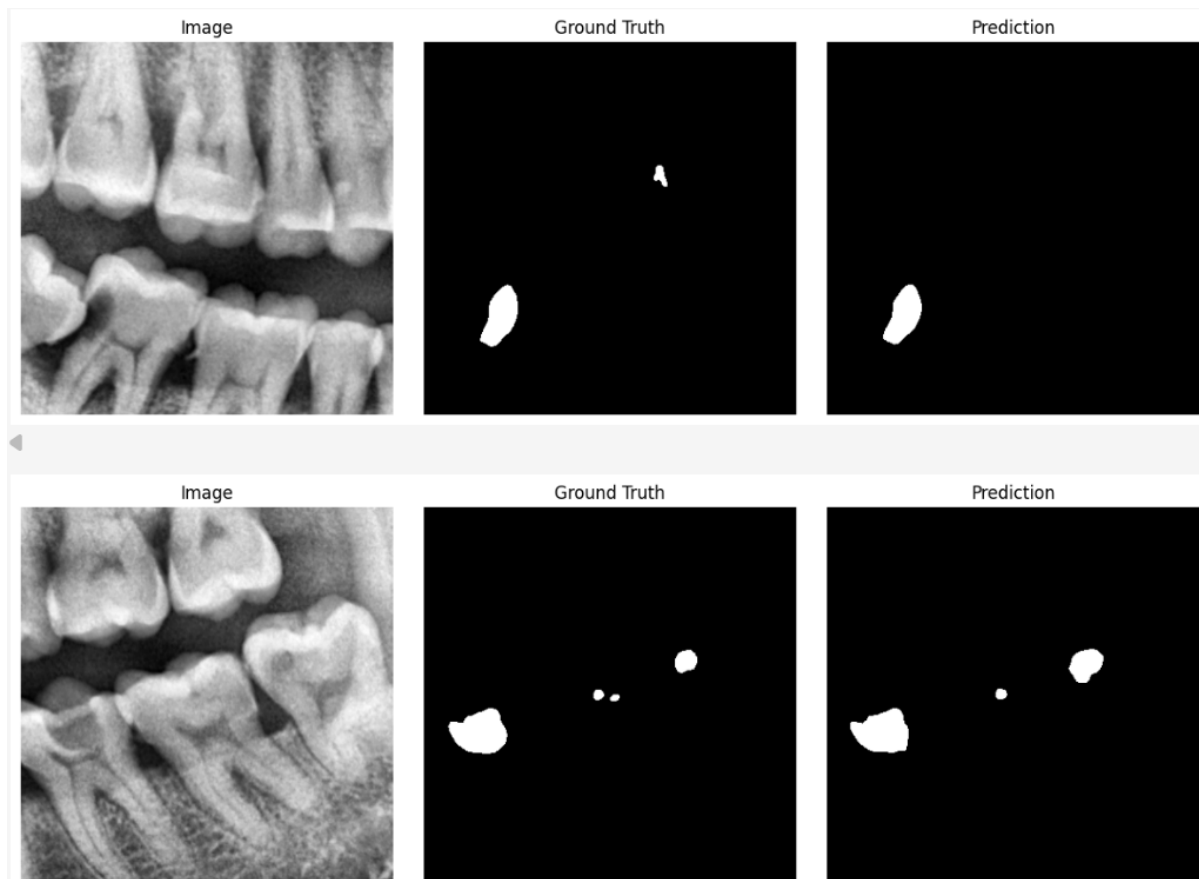


Figure 4: Qualitative segmentation results of the proposed MLUA model on the test set. Each row shows the input dental X-ray patch (left), ground truth annotation (middle), and predicted segmentation mask (right). Row 1 demonstrates successful localisation of small proximal caries lesions with minimal false positives. Row 2 shows correct detection of deep caries regions with well-preserved lesion boundaries, illustrating the model’s robustness to background artifact interference.

mized implementation reported in the original MLUA study, the proposed model attained higher sensitivity than the paper at both labeling regimes (64.57% vs. 58.77% for 265 slices, and 70.09% vs. 68.44% for 530 slices). This indicates that the model is particularly effective at identifying true lesion regions, which is of greater clinical importance than maximizing precision alone.

The reduced precision relative to the original implementation suggests that the model favors recall and may generate additional false-positive detections. From a clinical perspective, this trade-off is often acceptable because missing a lesion is generally more detrimental than flagging a suspicious region for further review.

Overall, the results confirm that the proposed Enhanced MLUA framework successfully reproduces the core semi-supervised learning principles of the original approach while introducing architectural innovations that improve lesion sensitivity and maintain competitive segmentation performance on the challenging DC1000 benchmark.

Table 7: Comparison with Existing Baselines on the DC1000 Dataset

Method	265 Labeled Slices			530 Labeled Slices		
	Dice (%)	SEN (%)	PRE (%)	Dice (%)	SEN (%)	PRE (%)
MAnet	40.37	39.37	54.81	65.03	61.06	75.34
FPN	43.43	41.90	55.62	64.66	59.30	76.30
DeepLabV3+	46.94	40.75	68.07	61.23	52.72	82.01
Linknet	45.10	47.11	53.50	66.08	61.47	77.07
Unet	42.51	43.68	53.89	64.99	63.07	72.74
Unet++	45.63	41.73	60.84	64.09	60.44	74.10
PAN	46.46	43.54	60.63	65.60	60.46	76.72
PSPNet	54.43	46.73	76.36	65.11	59.01	78.64
URPC	49.44	43.74	71.78	59.34	51.77	81.17
UAMT	52.31	52.90	59.14	67.79	64.60	76.02
CLCC	56.23	52.07	67.76	65.55	58.48	79.46
MLUA (Paper)	61.40	58.77	70.04	71.12	68.44	76.94
Proposed MLUA	58.71	64.57	61.05	60.58	70.09	60.28

5 Conclusion and Future Work

5.1 Summary of Contributions

This project successfully implemented and substantially extended the Multi-Level Uncertainty Aware (MLUA) semi-supervised framework for dental caries segmentation on panoramic radiographs. Starting from the core concepts introduced in the original MLUA methodology, a significantly enhanced architecture and training pipeline were developed directly from the proposed implementation.

The principal contributions of this work are summarized as follows:

1. **Complete PyTorch Lightning Implementation:** A full end-to-end implementation of the semi-supervised MLUA framework was developed using **PyTorch** and **PyTorch Lightning**, including data preprocessing, two-stream batch sampling, mean-teacher learning, Monte Carlo uncertainty estimation, dynamic certainty masking, and patch-based panoramic inference.
2. **Enhanced Segmentation Architecture:** The baseline FPN decoder was significantly extended with several advanced components, including Attention Gates, learnable Weighted Feature Merging, Convolutional Block Attention Module (CBAM), Atrous Spatial Pyramid Pooling (ASPP), deep supervision heads, and a dedicated boundary refinement branch.
3. **Hybrid Loss Function Design:** Multiple segmentation losses were implemented

and systematically evaluated, including BCE+Dice, Tversky, Focal Loss, Weighted BCE, and Vector Scaling Loss. The final hybrid objective combined these complementary losses to improve lesion sensitivity and robustness to severe class imbalance.

4. **Uncertainty-Aware Semi-Supervised Learning:** A complete mean-teacher framework was implemented in which uncertainty-filtered pseudo-targets were generated using Monte Carlo sampling with Gaussian noise and gamma augmentation. Only low-uncertainty regions contributed to the consistency objective, enabling effective exploitation of unlabeled data.
5. **Extensive Hyperparameter and Ablation Studies:** Systematic experiments were conducted to analyze the impact of Monte Carlo sampling counts ($T \in \{8, 10, 12\}$), training duration, consistency weights, loss functions, decoder variants, and inference thresholds on segmentation performance.
6. **Competitive Segmentation Performance:** The proposed Enhanced MLUA framework achieved a best validation Dice score of 0.6820, an IoU of 0.5740, and a sensitivity of 0.7637. In the standardized 265- and 530-labeled-slice experiments, the model achieved Dice scores of 58.71% and 60.58%, respectively, with sensitivity exceeding the original MLUA paper in both settings.
7. **Clinically Relevant Findings:** The experiments confirmed that emphasizing false-negative reduction through Tversky-based losses and uncertainty-guided consistency learning substantially improves lesion detection performance, which is critical for reducing missed caries diagnoses in clinical practice.

Overall, this project demonstrates that advanced architectural enhancements combined with uncertainty-aware semi-supervised learning can achieve strong and clinically meaningful performance while reducing dependence on costly pixel-level annotations.

5.2 Future Work

Several promising research directions can further improve the proposed framework:

- **Stronger Backbone Networks:** Replacing ResNet34 with more powerful encoders such as EfficientNet, ConvNeXt, or Swin Transformer may provide richer hierarchical representations and improved segmentation accuracy.
- **Pseudo-Label Curriculum Learning:** Combining uncertainty-filtered consistency loss with progressively refined pseudo-label supervision could further enhance the effective use of unlabeled data.

- **Multi-Class Caries Segmentation:** Extending the binary formulation (caries vs. background) to a three-class problem (shallow, middle, and deep caries) would provide clinically valuable severity assessment.
- **Full-Panorama Context Modeling:** Incorporating global attention mechanisms or hierarchical multi-scale architectures may better exploit long-range anatomical context across the entire panoramic radiograph.
- **Boundary-Aware and Topology Losses:** Additional contour-based or topology-preserving objectives may further improve lesion boundary delineation and reduce fragmented predictions.
- **Self-Supervised Pretraining:** Pretraining on large collections of unlabeled dental radiographs using contrastive or masked-image modeling techniques could improve feature quality prior to supervised fine-tuning.
- **Clinical Validation and Deployment:** Future work should evaluate the framework on multi-center datasets and integrate the model into computer-aided diagnosis systems for real-world dental screening applications.

In summary, the proposed Enhanced MLUA framework establishes a robust foundation for uncertainty-aware semi-supervised medical image segmentation and demonstrates the potential of advanced deep learning techniques to support efficient and reliable dental caries detection.

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